AWS SAA-C03 Exam Notes: Networking & VPC

1. Section Introduction

Networking is a significant portion of the Solutions Architect Associate exam. This section focuses on the Virtual Private Cloud (VPC), which acts as your isolated network environment within the AWS Cloud. Understanding how to secure, route, and connect VPCs is critical for passing.

- **Exam Tip:** Networking and Security combined often make up ~50% of the exam scenarios. Master VPC fundamentals early.

2. CIDR, Private vs Public IP

CIDR (Classless Inter-Domain Routing):

- Used to define IP ranges (e.g., `10.0.0.0/16`).
- **Slash notation (Subnet Mask):**
  - `/32` = 1 IP address.
  - `/24` = 256 IP addresses.
  - `/16` = 65,536 IP addresses.
- **VPC CIDR:** Minimum size `/28` (16 IPs), Maximum size `/16` (65,536 IPs).

Private vs Public IPs (RFC 1918):

- **Private IPs:** Only accessible within the network. Not routable on the internet.
  - `10.0.0.0` - `10.255.255.255` (Class A)
  - `172.16.0.0` - `172.31.255.255` (Class B)
  - `192.168.0.0` - `192.168.255.255` (Class C)
- **Public IPs:** Accessible from the internet. Unique across the web.
- **Exam Tip:** Memorize the RFC 1918 private ranges. If you see a question with a public IP (e.g., `54.x.x.x`) on a private database, it's a security flag.

3. Default VPC Overview

Every new AWS account comes with a Default VPC in every region.

- **Attributes:**
  - Has a CIDR block of `172.31.0.0/16`.
  - Has a Public Subnet in each Availability Zone (AZ).
  - Has an Internet Gateway attached.
  - Default Route Table routes `0.0.0.0/0` to the Internet Gateway.
  - **Use Case:** Getting started quickly.

- **Exam Tip:** For production scenarios or "secure architecture" questions, the answer is usually to create a **custom VPC**, not use the Default VPC.

4. VPC Overview

- **VPC (Virtual Private Cloud):** logically isolated portion of the AWS cloud.
- **Scope:** Region-bound (VPCs cannot span regions).
- **Limit:** Soft limit of 5 VPCs per Region per account.
- **Tenancy:**
  - **Default:** Shared hardware.
  - **Dedicated:** Dedicated hardware for your VPC (expensive).
- **Exam Tip:** Remember that VPCs belong to a specific **Region**, while Subnets belong to a specific **Availability Zone**.

5. Subnet Overview

- **Scope:** Availability Zone (AZ) bound (Subnets cannot span AZs).
- **AWS Reserved IPs:** AWS reserves the first 4 and the last 1 IP in *every* subnet CIDR block (5 total).
  - Example: `10.0.0.0` (Network), `10.0.0.1` (Router), `10.0.0.2` (DNS), `10.0.0.3` (Reserved), `10.0.0.255` (Broadcast).
- **Public Subnet:** Has a Route Table entry pointing `0.0.0.0/0` to an Internet Gateway (IGW).
- **Private Subnet:** Does *not* have a route to an IGW.
- **Exam Tip:** If a question asks how many usable IPs are in a `/28` subnet, the answer is $16 - 5 = 11$. Always subtract 5.

6. Internet Gateways (IGW) & Route Tables

Internet Gateway (IGW):

- Enables access to the internet for resources in a VPC.
- Horizontally scaled, redundant, and highly available (no bottleneck).
- **One** IGW per VPC.

Route Tables (RT):

- Contains rules (routes) to determine where network traffic is directed.
- **Main Route Table:** Automatically created.
- **Custom Route Table:** Can be created and associated with specific subnets.
- **Rule:** The most specific route matches (Longest Prefix Match).
- **Exam Tip:** You cannot attach more than one IGW to a VPC. If you need high availability for internet access, the IGW itself handles it internally; you don't need two.

7. Bastion Hosts (Jump Boxes)

- An EC2 instance in a **Public Subnet** used to SSH/RDP into private instances in a Private Subnet.
- **Security:**
  - Security Group should only allow SSH (port 22) from your corporate/home IP (not `0.0.0.0/0`).
  - Use Key Pairs securely.
- **Exam Tip:** If a scenario mentions "securely managing instances in a private subnet" and Session Manager isn't an option, select Bastion Host. Ensure the Security Group restricts ingress to a specific IP range (CIDR).

8. NAT Instances (Legacy)

- **Concept:** An EC2 instance launched in a public subnet to allow private instances to talk to the internet (outbound only).
- **Configuration:** Must disable **Source/Destination Check** on the instance.
- **Downsides:**
  - Not highly available out of the box.
  - Bandwidth limited by instance type.
  - User creates and manages security patches.
- **Exam Tip:** The critical step for NAT Instances is disabling the **Source/Destination Check**. Without this, it won't route traffic.

9. NAT Gateways

- **Concept:** AWS managed service for NAT (Network Address Translation). Allows outbound internet access for private subnets, but prevents inbound initiation.
- **Scope:** Created in a specific AZ.
- **Requirement:** Must be launched in a **Public Subnet** and requires an **Elastic IP (EIP)**.
- **High Availability:** For HA, create one NAT Gateway in each AZ.
- **Bandwidth:** Scales automatically up to 45 Gbps.
- **Cost:** Hourly charge + data processing charge.
- **Exam Tip:** If the question asks for "High Availability" for NAT, you must deploy one NAT Gateway in **each** Availability Zone. One NAT Gateway does not span multiple AZs.

10. NACL & Security Groups (Crucial Exam Topic)

Feature	Security Group (SG)	Network ACL (NACL)
Level	Instance Level	Subnet Level
State	**Stateful:** If you allow traffic IN, the return traffic is automatically allowed OUT.	**Stateless:** You must explicitly allow traffic IN and OUT.

Feature	Security Group (SG)	Network ACL (NACL)
Rules	Allow rules only. (Implicit deny at end).	Allow and Deny rules.
Order	All rules evaluated before decision.	Processed in number order (lowest first).
Scope	Applied to specific ENIs/Instances.	Applied to all instances in the Subnet.

- **Ephemeral Ports:** For NACLs, remember to allow return traffic on ephemeral ports (1024–65535).
- **Exam Tip:** If you need to **block** a specific bad IP address, you MUST use a **NACL**. Security Groups cannot explicitly deny traffic (they only "Allow").

11. VPC Peering

- Connects two VPCs using AWS private network.
- **Function:** Instances behave as if they are on the same network.
- **Rules:**
  - No overlapping CIDR blocks.
  - **Non-Transitive:** If A peers with B, and B peers with C, A cannot talk to C. (A needs to peer with C directly).
  - Works across Regions and Accounts.
  - You must update Route Tables in both VPCs.
- **Exam Tip:** Look out for "Transitive Peering" traps. A cannot talk to C through B. A must peer directly with C.

12. VPC Endpoints (PrivateLink)

Allows you to connect to AWS services using a private network (no Internet Gateway required).

1. **Interface Endpoints:**
   - Uses **ENI (Elastic Network Interface)** with a private IP.
   - Powered by PrivateLink.
   - Supports most AWS services (CloudWatch, SageMaker, etc.).
   - Costs money ($/hour + data).
2. **Gateway Endpoints:**
   - Uses a target in the **Route Table**.
   - Supports only **S3** and **DynamoDB**.
   - **Free** of charge.

- **Exam Tip:** If the question mentions "S3" or "DynamoDB" specifically, look for **Gateway Endpoint**. For almost any other AWS service (like Kinesis, CloudWatch), look for

Interface Endpoint.

13. VPC Flow Logs

- Captures information about the IP traffic going to and from network interfaces.
- **Levels:** VPC, Subnet, or ENI level.
- **Destinations:** S3, CloudWatch Logs, or Kinesis Data Firehose.
- **Usage:** Troubleshooting connectivity, security monitoring.
- **Note:** It captures metadata (Source IP, Dest IP, Port, Action), not the actual content of the packet.
- **Exam Tip:** Flow Logs do **not** capture the packet content (payload). If the question asks to inspect the *content* (e.g., for malware signature), Flow Logs is the wrong answer (use Traffic Mirroring or Network Firewall).

14. Site-to-Site VPN, Virtual Private Gateway & Customer Gateway

Connects On-Premises data center to AWS VPC over the **public internet**.

- **Virtual Private Gateway (VGW):** The VPN concentrator on the **AWS side**.
- **Customer Gateway (CGW):** Physical device or software application on the **Customer (On-prem) side**.
- **Site-to-Site VPN:** The IPsec encrypted tunnel connecting VGW and CGW.
- **Setup:** Quick to set up, but latency is dependent on internet conditions.
- **Exam Tip:** The VGW is attached to the VPC. The CGW represents the physical device at the customer's office. Don't mix them up.

15. AWS VPN CloudHub

- **Concept:** Allows you to securely communicate from one remote site to another using the AWS VPN infrastructure.
- **Architecture:** Operates on a **Hub-and-Spoke** model where the AWS VPC (specifically the Virtual Private Gateway) acts as the Hub.
- **Requirement:** Multiple Customer Gateways (one for each remote site) must connect to the *same* Virtual Private Gateway (VGW).
- **Routing:** Uses **BGP** (Border Gateway Protocol). Each Customer Gateway must use a unique BGP Autonomous System Number (ASN).
- **Cost:** Low cost; leverages the public internet.
- **Exam Tip:** If a scenario describes multiple branch offices that need to talk to each other AND to AWS, and asks for a low-cost solution (avoiding MPLS/Direct Connect), choose **VPN CloudHub**.

16. Direct Connect (DX) & Direct Connect Gateway

Connects On-Premises to AWS via a **dedicated private physical fiber connection**.

- **Benefits:** Consistent network performance, high bandwidth, lower data egress rates.
- **Security:** Traffic is private but **not encrypted** by default.
- **Direct Connect Gateway:** Allows a Direct Connect connection to interface with VPCs in **multiple regions**.
- **Setup:** Takes weeks/months (physical cabling required).
- **Exam Tip:** Direct Connect is **not** encrypted by default. If the requirements are "Dedicated connection" AND "Encryption", you need DX + VPN.

17. Direct Connect + Site-to-Site VPN

- If you need a **dedicated connection** (Direct Connect) AND **encryption** (VPN).
- You establish a public Virtual Interface (VIF) on Direct Connect, then create an IPsec VPN tunnel over that connection.
- Result: Secure, encrypted, dedicated, high-bandwidth connection.
- **Exam Tip:** This is the standard answer for "High security compliance (encryption)" + "High bandwidth/reliability".

18. Transit Gateway (TGW)

- **Hub-and-Spoke model:** Connects thousands of VPCs and on-prem networks.
- **Transitive Routing:** Unlike peering, TGW allows transitive routing (A ↔ TGW ↔ B).
- **Cross-Region:** Can peer TGWs across regions.
- **Complexity:** Reduces complexity of mesh peering networks.
- **Multicast:** TGW is the only service that supports IP Multicast.
- **Exam Tip:** If the question mentions "simplifying network topology" or connecting "hundreds/thousands of VPCs," the answer is **Transit Gateway**.

19. VPC Traffic Mirroring

- Allows you to capture and inspect network traffic.
- **Mechanism:** Copies traffic from an elastic network interface (Source) and sends it to a target (Target).
- **Target:** Network Load Balancer (NLB) or another ENI.
- **Use Case:** Intrusion Detection Systems (IDS), packet inspection, compliance monitoring.
- **Exam Tip:** This is used for "Deep Packet Inspection" or analyzing actual payloads for security threats, which Flow Logs cannot do.

20. IPv6 for VPC

- VPCs can operate in Dual Stack mode (IPv4 and IPv6).
- **Addressing:** IPv6 addresses are globally unique (Public). There are no private IPv6 ranges in AWS VPC context.

- **OSI:** Every instance gets a global Unicast IPv6 address.
- **Exam Tip:** You cannot disable IPv4 for a VPC (yet); you enable IPv6 *in addition* to IPv4 (Dual Stack).

21. Egress Only Internet Gateway

- **IPv6 Only.**
- Similar to a NAT Gateway but for IPv6.
- Allows instances to initiate outbound connections to the internet over IPv6 but **prevents inbound** connections.
- **Update Route Table:** Route `::/0` to the Egress-Only Internet Gateway.
- **Exam Tip:** Egress Only Internet Gateway is **exclusive to IPv6**. Do not pick it for IPv4 scenarios.

22. VPC Section Summary

- **Security:** Use NACLs for stateless subnet blocking, SGs for stateful instance allowance.
- **Connectivity:** Peering for simple 1:1, Transit Gateway for complex Many:Many.
- **Hybrid:** VPN for quick/encrypted, Direct Connect for reliable/high-throughput.
- **Access:** Bastion for SSH, Endpoints for internal AWS service access.
- **Outbound:** NAT Gateway (IPv4) / Egress-Only IGW (IPv6).
- **Exam Tip:** Always verify if the traffic needs to be encrypted. DX is private but unencrypted. VPN is public but encrypted.

23. Networking Costs in AWS

- **Inbound Data:** Generally free.
- **Outbound Data:** Costs money.
- **Private IP communication:** Generally cheaper than Public IP.
- **NAT Gateway:** Hourly charge + Per GB data processing fee (can get expensive).
- **VPC Endpoints:** Interface endpoints cost hourly; Gateway endpoints are free.
- **Data Transfer between AZs:** Costs money (if using public IPs or peering).
- **Exam Tip:** Using Private IPs to transfer data between instances in the same region is usually the most cost-effective method.

24. AWS Network Firewall

- Managed, stateful, network firewall and intrusion detection/prevention service (IDS/IPS).
- **OSI Layer:** Protects Layers 3-7.
- **Capabilities:**
  - Web filtering (domain names).
  - Inspection of VPC traffic.
  - Compatible with "Suricata" rules.

- **Deployment:** Deployed using a "Firewall Subnet" in the VPC. Route tables must be updated to send traffic through the firewall endpoint.
- **Exam Tip:** If a question asks for "Domain Name filtering" (e.g., block `facebook.com` for corporate instances) across the entire VPC, AWS Network Firewall is a strong candidate (DNS Firewall is another, but Network Firewall is broader).